

KUG: Joint Enhancement of Internal and External Knowledge for Retrieval-Augmented Generation

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Abstract

Query enhancement, a pivotal methodology in Retrieval-Augmented Generation (RAG) for addressing information scarcity in queries, has garnered increasing research attention. Nevertheless, existing approaches overlook the inherent distinctions between domain-specific knowledge and external factual sources during integration. To bridge this gap, we propose KUG (Knowledge-Update-Generation), a novel RAG framework that leverages internal knowledge semantics to ensure query enhancement efficacy, validates and dynamically updates knowledge representations using external evidence, and achieves systematic integration through knowledge graph embeddings. Extensive experiments on six standard BEIR benchmarks demonstrate that KUG outperforms the state-of-the-art methods, achieving an improvement of 1%–2% in recall metrics. Notably, the framework demonstrates significant performance gains in multi-hop reasoning tasks, advancing the development paradigm for RAG systems. The code will be public soon.

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CCS Concepts

• **Information systems** → **Query reformulation**; *Query suggestion*; *Query intent*; Retrieval effectiveness; Recommender systems; Document filtering; • **Computing methodologies** → **Natural language processing**; Search methodologies.

Keywords

Retrieval Augmented Generation, Query Enhancement, Large Language Model

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1 Introduction

Retrieval Augmented Generation (RAG) has emerged as a prominent research focus in recent years [4, 11, 20]. Current mainstream retrieval methodologies predominantly focus on measuring the textual similarity between queries and documents. However, in practical applications, the textual similarity between query statements and target gold-standard documents often remains suboptimal. In particular, when the query content exhibits excessive brevity or informational paucity, the difficulty of accurately identifying target documents increases substantially [1, 15–17]. In this context, query optimization techniques have progressively evolved into a focal area of academic research and technological practice. Among these,

query enhancement technology prioritizes information augmentation as its core objective, aiming to bridge the semantic gap between queries and documents through the integration of additional relevant information into query statements.

The investigation of information sources for query enhancement originates primarily from two distinct dimensions: knowledge generated from the internal parameters of LLMs and retrieved content from external knowledge bases. Early scholarly efforts focused predominantly on independent exploration of these two paradigms. The former approach capitalizes on the inherent parametric knowledge to enrich user queries with supplementary information or generate auxiliary reference answers[33, 37]. The latter methodology retrieves relevant content from external repositories to address query-related information gaps[7]. Recent academic advancements have witnessed an emerging paradigm shift toward synergistic integration of both approaches, leading to increased methodological sophistication in query enhancement techniques. Contemporary hybrid methods simultaneously leverage dual information sources while performing augmentation operations. However, most existing integration mechanisms adopt equal fusion strategies, treating the two merely as knowledge sources of the same nature without fully considering the essential characteristic differences between the two types of information sources[8, 13].

Specifically, LLM-generated knowledge typically demonstrates greater relevance to query resolution but carries potential hallucination risks. In contrast, retrieval of external knowledge base offers enhanced content authenticity, but suffers from limited alignment with query objectives – a fundamental constraint inherent in retrieval-difficult scenarios[6]. The simple fusion of these two heterogeneous information sources does not fully exploit their potential, leading to retrieval accuracy below expectations. This contradiction raises a critical research challenge: How to organically integrate the relevance advantages of internal model knowledge with the authenticity guarantees of external retrieval, thereby simultaneously mitigating hallucinations and irrelevant information interference? Addressing this challenge has become a frontier research direction with significant theoretical value.

To this end, this study breaks through the inherent limitations of the traditional retrieval-generative paradigm and proposes a novel query enhancement framework, KUG (Knowledge Update Generate). The framework takes information update as the core component, acquiring internal parameterized knowledge and external knowledge base knowledge, respectively, using low-confidence high-relevance internal knowledge as retrieval guidance and high-confidence low-relevance external knowledge as verification basis. Through a triplet-based information integration strategy, it constructs an information verification and update mechanism, effectively improving the performance of general retrieval tasks, achieving complementary advantages of the two information sources, and avoiding their inherent defects.

Specifically, the KUG framework adopts a four-stage collaborative process: in the Knowledge Stage, it utilizes the reasoning capabilities of LLMs to acquire internal knowledge and employs a Retriever to retrieve external knowledge from external knowledge bases; in the Update Stage, it aligns retrieval results with LLM-generated content by constructing a dynamic knowledge graph, identifies and replaces potential hallucinated content, and augments

knowledge; in the Generation Stage, it uses the updated knowledge graph to generate enhanced documents with high-confidence information; and in the Final Retrieval Stage, it performs re-retrieval by combining the original query with enhanced documents and outputs results.

The framework achieves significant performance improvements across multiple datasets from the BEIR benchmark, with an average boost of 1%-2%, particularly excelling in multi-hop reasoning tasks. Experiments demonstrate that the structured update mechanism of the dynamic knowledge graph effectively mitigates semantic drift during LLM generation processes, enhancing the relevance and factual consistency of the retrieval results. Through in-depth analysis of the experimental data, we found that the KUG framework exhibits excellent discriminative capabilities for questions with similar query semantics but completely heterogeneous answer logic. To validate this characteristic, this study constructed a small-scale validation dataset FIQA-GR focused on the scenario of similar query semantics but heterogeneous answer logic, based on the FIQA dataset from the BEIR benchmark. Experimental results show that on the FIQA-GR dataset, the recall of the KUG framework significantly outperforms related work, thus verifying the framework's effectiveness in such tasks.

The contributions of this work are as follows.

- We introduce a novel KUG (Knowledge-Update-Generation) framework that integrates model parameters from LLMs with external knowledge bases. This iterative mechanism combines the complementary strengths of both sources of knowledge while mitigating their respective limitations.
- Through a detailed analysis of data characteristics, we constructed a small-scale validation dataset named FIQA-GR, focusing on scenarios involving semantically similar queries with heterogeneous reasoning mechanisms.
- Extensive experiments on the BEIR benchmark dataset demonstrate the generalizability of the proposed framework across diverse retrieval tasks. The results show significant improvements in the relevance and applicability of the retrieved information compared to state-of-the-art baselines.

This paper is organized in following sections: Section 2 describes related work in this area along with some background on Query Enhancement. Section 3 describes our approach in detail. Section 4 describes our experimental settings and discuss the findings from our empirical study. In Section 5 we conclude our work. Finally, in Section 6 we show our limitations and future. All code as well as detailed instructions are available at <https://github.com/LMY-FDU/KUG>.

2 Related Work

2.1 Retrieval-augmented Generation

Due to their inherent characteristics, LLMs often face many serious challenges in practical applications, especially when they are unable to accurately answer certain questions. The model may generate incorrect information, a phenomenon known as "hallucination". Research has shown that one of the main causes of hallucinations is the lack of relevant knowledge within the model, and enhancing external knowledge often requires retraining, which is not only costly but also slow in knowledge updates[11]. Therefore,

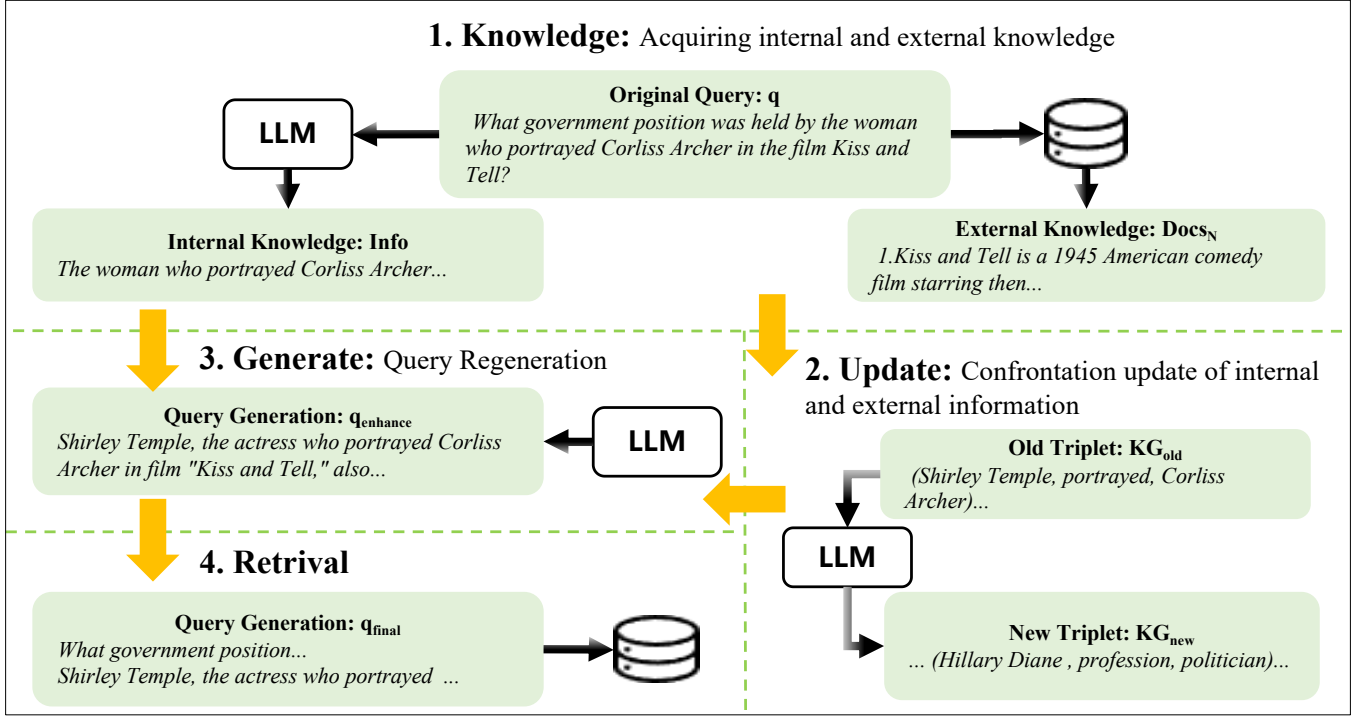


Figure 1: Method overview of KUG: The architecture diagram of the KUG framework illustrates its distinctive Generation-Retrieval-Update-Generation-Retrieval process. The framework comprehensively leverages the advantages of both internal and external knowledge. Through this process, it achieves effective collaboration between internal and external information, demonstrating robust performance in information retrieval and generation tasks.

Retrieval Augmented Generation (RAG) has emerged as a key research direction[20]. In recent years, RAG research has exhibited a diversified development trend[28]. Additionally, some studies have attempted to integrate RAG with knowledge graphs, leveraging structured information from these graphs to further enhance model reasoning capabilities and effectively reduce the frequency of hallucinations[26].

2.2 Query Enhancement

With the continuous innovation in Retrieval-Augmented Generation (RAG) technology, researchers have proposed a series of improved strategies to enhance its effectiveness. However, retrievers in current RAG frameworks still face critical challenges that require urgent attention, significantly degrading the efficiency and accuracy of retrieval systems. Consequently, researchers and practitioners have conducted extensive investigations into query optimization techniques. Building on existing research, these techniques can be systematically categorized into four main directions: Query Expansion, Question Decomposition, Query Disambiguation, and Query Abstraction[2, 30, 34]. Query Expansion focuses on addressing information insufficiency in queries by integrating external knowledge sources to enrich the original query, thereby expanding the retrieval scope and improving the comprehensiveness of results.

From the dimension of information gain sources, the core implementation paths of query enhancement can be divided into two

categories: one is knowledge generated by internal parameters of large language models (LLMs)[33, 37], which generates supplementary information semantically related to queries by activating the knowledge distribution within LLMs; the other is the fusion of exogenous knowledge achieved by retrieving external knowledge sources, specifically including explicit knowledge carriers such as knowledge graphs, large-scale corpora, and pre-retrieved document sets[7].

In research on LLM-based query enhancement, HYDE is one of the early representative studies that leveraged LLM to generate potential answers and directly replace original queries for retrieval[9]. However, it faced significant bottlenecks in performance improvement. In contrast, the BLADE study used domain-specific models to generate auxiliary drafts, which effectively enhanced collaborative alignment between models and retrievers, but still lacked sufficient mechanisms to prevent and handle hallucinations[21]. MEMORAG constructs multiple latent clues for queries to support the retrieval process, yet it is constrained by the context length limitations of LLMs[27]. Query enhancement strategies based on reinforcement learning (such as DeepRetrieval) adaptively optimize retrieval performance through dynamic interaction with retrieval systems, but suffer from high training costs and insufficient generalization capabilities[14]. In the application of external knowledge retrieval, acquiring information from external knowledge bases has become a crucial means of query enhancement. Many studies have utilized external knowledge to perform semantic completion on

query statements, significantly improving the matching between search results and user needs. However, this type of method is prone to introducing irrelevant information and interfering with documents in practical applications.

Furthermore, some research attempts to integrate LLM-generated content with externally pre-retrieved information, optimizing retrieval results through comprehensive ranking strategies[8, 13]. However, existing studies neither systematically analyzed the complementary strengths and limitations of these two information sources[19] nor proposed adaptive processing schemes tailored to their unique characteristics. The ReFeed method innovatively generates multiple candidate responses using an LLM, retrieves relevant documents, reranks the retrieved documents, and selects the most contextually aligned document as the final retrieval result[38]. This gap highlights opportunities for future research to explore principled frameworks that leverage the distinct advantages of internal model knowledge and external knowledge bases.

3 KUG

3.1 Overall Framework

The KUG framework comprises three key steps: acquiring internal and external knowledge, adversarially updating internal and external information, and query regeneration (as shown in Figure 1). First, in the information acquisition stage, the LLM and retriever are employed to obtain internal parametric knowledge and external knowledge, respectively. Analysis shows that while the LLM may generate hallucinations, it possesses reasoning capabilities and can generate information helpful for answering queries. In contrast, external knowledge resources often contain noisy or irrelevant documents but are generally factually accurate; this distinction motivates subsequent steps: leveraging internal knowledge as a basis and external knowledge to mitigate hallucinations. To reduce interference from internal knowledge hallucinations with accurate external knowledge, the framework replaces flawed internal knowledge and supplements information gaps—potentially arising from the model’s limited exposure to the dataset—with external knowledge. The core step of this process, “adversarial update of internal and external information,” optimizes internal knowledge through strategic replacement and supplementation using external knowledge. Finally, in the query regeneration stage, the LLM restructures knowledge according to the semantic requirements of the original query. The retrieval system then processes a hybrid input combining the original query with the re-generated context, synergistically leveraging the complementary strengths of internal and external knowledge to mitigate LLM hallucinations and external data noise.

3.2 Framework Process

3.2.1 Acquiring Internal and External Knowledge. As the foundational step of the framework, this process primarily involves the LLM performing semantic interpretation on the original query to generate semantically relevant internal knowledge. This internal knowledge serves as a basis for relevant knowledge augmentation, ensuring the framework’s high-relevance advantage. Subsequently, by concatenating the internal knowledge with the original query, a Dense Passage Retrieval (DPR) approach is employed to perform

semantic matching against the external knowledge base. This step retrieves high-confidence external knowledge that ensures factual consistency, providing factual support for subsequent adversarial training-based knowledge refinement and guaranteeing the framework’s high-accuracy advantage.

$$Info = LLM(q) \quad (1)$$

In this paper, the symbol q denotes the original query, while the symbol $Info$ is used to refer to the internal knowledge generated by LLM.

$$Docs_N = \text{Retriever}(q, Info, Docs, N) \quad (2)$$

Here, $Docs$ denotes the documents to be retrieved, and $Retriever$ refers to the retriever (mainstream retriever methods include sparse retrieval and dense retrieval, among others)[15]. $Docs_N$ represents the set of documents obtained after preliminary retrieval, from which the $topN$ documents are selected as reference documents.

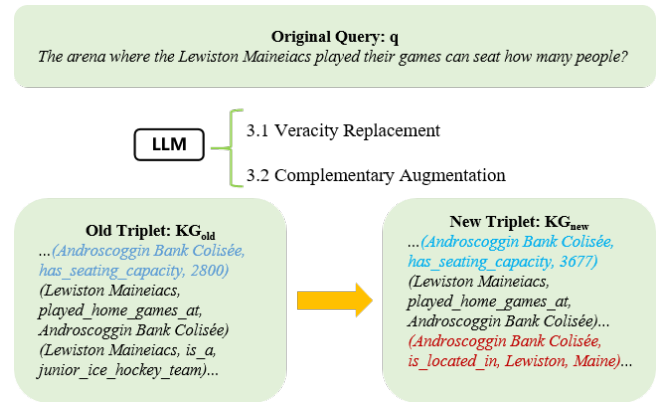


Figure 2: The schematic diagram of the Confrontation Update of Internal and External Information stage shows that the blue text is labeled as "authenticity replacement" and the red text is labeled as "supplementary enhancement".

3.2.2 Updating Internal and External Information. On the basis of integrating internal and external knowledge, this stage calibrates and enhances knowledge representation by designing a collaborative update mechanism. This mechanism leverages internal knowledge as the semantic backbone and external knowledge as factual grounding, systematically optimizing the relevance and accuracy of knowledge representation through a triple-based knowledge graph system. Specifically, this study represents internal knowledge as a knowledge graph composed of triple sets and uses external knowledge to update the graph in two ways: (1) authenticity replacement, which identifies information in external knowledge conflicting with internal knowledge (e.g., factual contradictions or logical inconsistencies) and replaces such information to ensure factual accuracy; (2) supplementary enhancement, which leverages the reasoning capabilities of LLMs to autonomously select information from external knowledge that is absent in internal knowledge but enhances retrieval relevance (e.g., supplementing entity attributes or expanding relationship types). This update strategy retains the deep semantic understanding of queries from internal knowledge while

correcting potential hallucinations or errors using the authenticity of external knowledge, thereby achieving synergistic optimization of relevance and factual consistency[12, 24]. As shown in Figure

Prompt for q_{enhance} :

Provide a knowledge graph consisting of triplets for a query in the RAG system, and generate a document from the triplets that can assist in answering. The length must not exceed 256 tokens. Output format requirements: <DOC>(generated doc)<DOC> Enter as follows:...

Figure 3: This figure shows the prompt used in the query regeneration phase. To ensure efficiency, the maximum length of the prompt is set to 256 tokens.

2, the specific example illustrated demonstrates the critical role and effectiveness of this stage. The framework effectively identifies and corrects erroneous information in internal knowledge while performing supplementation and enrichment of external knowledge, thus addressing the issues of knowledge hallucinations and information gaps.

In the current framework design, this study employs an LLM to implement knowledge extraction and normalization. Notably, this functional module can also be implemented using traditional NLP techniques, which demonstrates both the flexibility and scalability of the methodological system and the framework’s potential for adaptability and application scalability.

$$KG_{\text{old}} = LLM(Info) \quad (3)$$

$$KG_{\text{new}} = LLM(q, KG_{\text{old}}, Docs_N) \quad (4)$$

Herein, KG_{old} denotes the set of triples generated from $Info$ through the model reasoning process. KG_{new} represents the set of triples that integrates internal and external knowledge, generated based on the original query q , KG_{old} , and $Docs_N$ (as defined above).

3.2.3 Query Regeneration. Currently, structural reconstruction of the triplet set is required to generate query-oriented retrieval documents. Although the adversarial update mechanism yields an integrated set of internal and external knowledge, the original triplet structure lacks query-specific semantic alignment. To address this limitation, the framework employs an end-to-end query regeneration strategy: taking the original query q and the structured triplet set as input, it leverages the semantic parsing capability of LLMs to generate an enhanced query q_{enhance} . This enhanced query retains the core intent of the original query while incorporating semantic constraints from structured knowledge. Finally, the enhanced query is concatenated with the original query as q_{final} , thereby forming a multi-dimensional semantic representation to guide the final retrieval process.

$$q_{\text{enhance}} = LLM(q, KG_{\text{new}}) \quad (5)$$

$$q_{\text{final}} = [q; q_{\text{enhance}}] \quad (6)$$

Herein, q_{enhance} denotes the final enhanced query integrating internal and external information, while q_{final} represents the collection of documents used for the final retrieval.

4 Evaluation

4.1 Baselines

In query enhancement research, we establish baseline retrieval using the BGE model[35] and select representative methods from single-source knowledge query enhancement approaches: HYDE[10] and Query2doc[33], along with the state-of-the-art LameR model[29] that integrates dual information sources as comparative baselines. Specifically, HYDE improves retrieval performance by generating hypothetical answer representations, while Query2doc enhances retrieval through few-shot demonstration-based query expansion. LameR implements a collaborative query enhancement strategy integrating internal and external knowledge: conducting initial retrieval on raw queries followed by iterative refinement based on retrieved documents, establishing an optimized information fusion paradigm.

4.2 Datasets and Metrics

4.2.1 Datasets. As a new-generation benchmark for evaluating information retrieval models in cross-domain scenarios, the Benchmark for Evaluation of Information Retrieval Models in Cross-Domain Scenarios (BEIR) provides a standardized framework for assessing the performance of retriever in complex environments[31]. Leveraging BEIR’s cross-domain evaluation capabilities, this study constructs a subset including the following datasets: HotpotQA (multi-hop question answering)[36], FIQA (financial domain)[25], NRCorpus (medical information retrieval)[3], SciFact (scientific fact verification)[32], NQ (Natural Questions open-domain QA)[18], and Quora (duplicate question retrieval)[5]. These datasets cover multiple types of task scenarios, enabling quantitative evaluation of various capabilities of the framework and possessing high persuasiveness.

Given the relatively large sizes of the HotpotQA, NQ, and Quora corpora and their respective test sets, and constrained by the computational resources and time limitations of the experimental setup, the following measures were adopted: 200 queries were randomly selected from each dataset to form a test subset, and the corpus document count was initially capped at 60,000 documents. However, preliminary experiments revealed unexpectedly high recall rates for NQ and Quora, necessitating further expansion of their corpus sizes to 100,000 documents to enable more comprehensive model evaluation.

4.2.2 Metrics. In the main experiment of this work, the evaluation metric primarily adopted is recall. The reason for choosing recall over ranking-sensitive metrics is mainly determined by the characteristics of the application scenario. The core value of recall lies in measuring the system’s ability to comprehensively capture potentially relevant information, while ranking-sensitive metrics essentially focus on ranking precision. With the continuous improvement of large language model capabilities, their ability to accurately locate target content in massive information (i.e., the "needle-in-a-haystack" capability) has significantly enhanced[22]. When the model does not strictly limit the number of candidate documents, the core task of the retriever shifts to recalling relevant knowledge as much as possible rather than finely ranking a limited number of candidates. Therefore, when the k -value is within

Table 1: In the main experiment, recall@40 was adopted as the evaluation metric to quantify the framework’s capabilities, and the method proposed in this study achieved optimal performance on all four sub-datasets while surpassing existing research results in terms of average scores.

Method	NFCorpus	FIQA	FIQA-GR	HotpotQA	SciFact	NQ	Quora	Avg
BGE	0.2987	0.6776	<u>0.7678</u>	0.93	0.9172	<u>0.9525</u>	0.9883	<u>0.7909</u>
HyDE	<u>0.3079</u>	0.6736	<u>0.7581</u>	0.8825	<u>0.9483</u>	0.9225	0.9680	0.7801
Query2doc	0.2882	0.6786	0.7567	0.8925	0.945	0.9225	0.9824	0.7809
LameR	0.2939	<u>0.6788</u>	0.7637	<u>0.9475</u>	0.8988	0.935	<u>0.9874</u>	0.7868
KUG(ours)	0.3098	0.6846	0.7899	0.9575	0.931	0.9575	0.9839	0.8027

an appropriate range, recall is sufficient to reflect the retriever’s capability and positively impact the subsequent generation stage. However, considering that ranking-sensitive metrics are commonly used and widely applied in performance evaluations of existing related studies, this research will select typical ranking-sensitive metrics such as Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP) as evaluation metrics in the hyperparameter experiments and ablation experiments to comprehensively assess the impact of various hyperparameter configurations and different modular components on the framework’s performance.

4.3 FIQA-GR

Experimental results demonstrate that the KUG framework effectively mitigates semantic query ambiguity—scenarios where queries exhibit high semantic similarity but differ only in key entities, temporal attributes, or locations. To rigorously evaluate this capability, we constructed a specialized validation dataset, FIQA-GR, based on the FIQA corpus. Constrained by annotation resources, the study first selected 200 question-answer pairs from the original FIQA dataset. Leveraging the Qwen-3 model, we generated 200 additional QA pairs with high semantic similarity to the original queries but featuring entity or attribute substitutions, yielding a test set of 400 queries. The FIQA-GR dataset integrates these synthetic documents with the original FIQA corpus, resulting in a comprehensive experimental dataset comprising 58,172 documents. This dataset focuses explicitly on evaluating models’ ability to discriminate between semantically similar queries with heterogeneous key information, thereby enabling targeted validation of the KUG framework’s efficacy in complex semantic contexts.

To clearly demonstrate the characteristics of this dataset, the following examples are provided:

FIQA original query: *Can a entrepreneur hire a self-employed business owner?* Corresponding answer: *Yes. I can by all means start my own company and name myself CEO. If Bill Gates wanted to hire me, I’ll take the offer and still be CEO of my own company. Now, whether or not my company makes money and survives is another question. This is the basis of self-employed individuals who contract out their services.*

New query generated based on the original query: *Can a freelancer hire a self-employed consultant?* Corresponding answer: *Freelancers often rely on self-employed professionals for specific tasks. This is common in industries like IT, marketing, and design, where individuals offer their skills on a project-by-project basis.*

4.4 Hyper-parameters

This study employs the Qwen-2-7b model, which is built upon extensive multimodal training data and features tens of billions of parameters, as the sole generative component. This model demonstrates exceptional performance in natural language generation, logical reasoning, and knowledge integration tasks. To ensure consistency across experimental conditions, all comparative methods utilize the same model. In the pre-retrieval configuration, the number of documents is uniformly set to 2. For the comparative LameR model, the pre-retrieval document count is similarly constrained to 2 to maintain a fair basis for comparison. The prompts designed in the original LameR paper are directly adopted without additional optimization. The specific experimental configurations are as follows: The temperature parameter is set to 0 (Temperature=0) to ensure deterministic output. The retrieval module employs ByteDance’s BGE-large-en-v1.5, a 150-million-parameter cross-lingual semantic encoding model optimized via contrastive learning to generate deep semantic representations, as the core retriever. This model, belonging to the dense retrieval paradigm, demonstrates superior recall and precision in long-text semantic matching and cross-lingual retrieval tasks. The final number of retrieved documents is set to 40 (topk=40). These parameters establish the experimental settings, ensuring equitable comparison among different methods under identical conditions.

4.5 Main Experiments

As shown in Table 1, under identical hyperparameter settings, the proposed KUG framework demonstrates superior performance across four benchmark datasets, demonstrating its effectiveness in knowledge-enhanced reasoning tasks. Specifically, compared with baseline models, the framework achieves an average accuracy improvement of 1.0 percentage points across these evaluation sets. In research benchmarks, such statistically significant gains not only reflect the architectural advantages of our framework but also underscore its practical applicability in real-world scenarios. Notably, the framework demonstrates particularly strong performance on both the established HotpotQA multi-hop QA benchmark and the recently introduced FIQA-GR dataset. This enhanced capability stems from the novel knowledge integration mechanism: through adaptive fusion of model-inherent knowledge with external domain-specific information, the framework establishes optimized reasoning pathways for complex queries, consequently improving

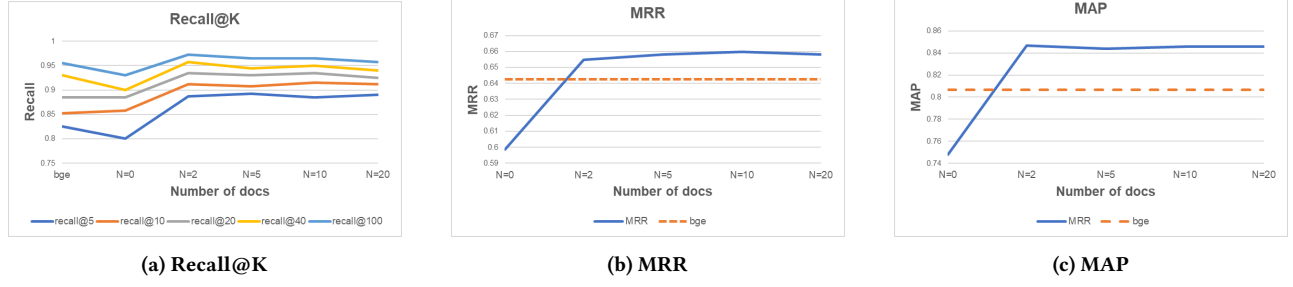


Figure 4: The figure illustrates the relationship between the number of pre-retrieval documents, denoted as N , and the performance of the framework. When investigating the impact of the number of reference documents on performance, when $N = 0$, we use "have no new docs." as a substitute.

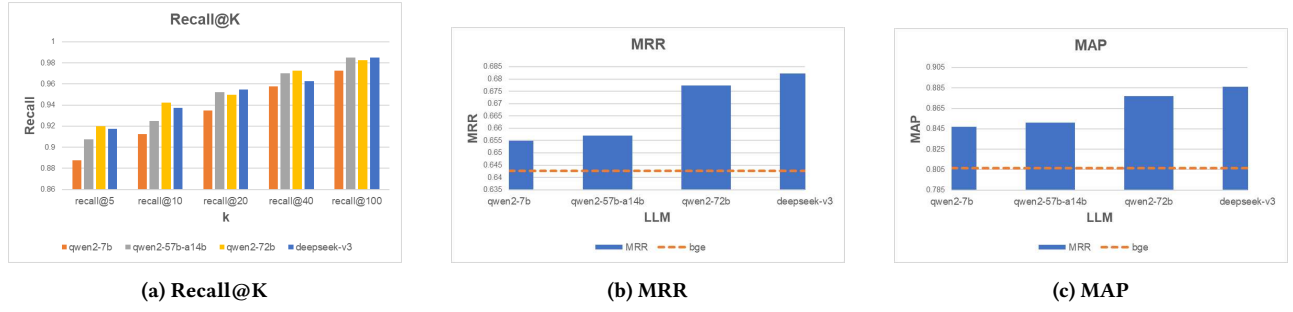


Figure 5: The framework’s performance metrics (recall@k, MRR, MAP) exhibit statistically significant positive correlations with LLM pre-training scale. With increasing model parameters, consistent monotonic improvements are observed across all three metrics, while maintaining robust cross-architecture compatibility with diverse LLM implementations.

the model’s multi-step logical reasoning capacity in challenging QA scenarios.

It is worth noting that on the SCIFACT scientific fact-checking dataset and the Quora duplicate question retrieval dataset, the performance of the KUG framework is slightly inferior to HyDE and BGE, respectively. On the NFCorpus medical dataset, the performance improvement is relatively limited. In-depth analysis reveals that the performance differences on the NFCorpus and SCIFACT datasets may stem from the limitations of the Qwen-2-7B base model in understanding professional knowledge in the scientific domain. When processing scientific literature, the model struggles to accurately discern complex logical relationships, introducing semantic biases during the construction of triple knowledge units. Specifically, tasks involving specialized scientific terminology, experimental methodology descriptions, or cross-literature data associations significantly degrade the entity relevance and relationship accuracy of generated triples, thereby undermining the effectiveness of knowledge alignment in the verification stage. To address this, future research could fine-tune the LLM with domain-specific knowledge or integrate professional knowledge graphs to enhance triple generation quality and improve the framework’s retrieval performance in specific domains. Regarding the performance disparity on Quora, it stems from a domain mismatch: Quora’s retrieval objective is duplicate queries rather than documents, whereas KUG framework prompts focus on target documents, inadvertently incorporating extraneous information that reduces effectiveness.

4.6 Number of Reference Documents

To systematically evaluate the optimal performance envelope of the KUG framework, we conducted rigorous hyperparameter ablation studies on the HotpotQA dataset, a benchmark that requires intricate multi-step reasoning and precise document correlation analysis through retrieval systems. This dataset is uniquely suited for assessing parameter sensitivity in complex reasoning scenarios. We systematically analyzed the impact of key framework parameters on both recall-focused and ranking-focused evaluation metrics—including mean reciprocal rank (MRR) and mean average precision (MAP)—which collectively quantify retrieval ranking performance. Importantly, the retriever configuration (specifically the encoder architecture) was held constant across all experiments, with the BGE-large-en-v1.5 model consistently serving as the fixed retrieval component to maintain methodological consistency throughout the investigation.

In the "Internal-External Information Verification" module, the number of pre-retrieved documents N serves as a critical hyperparameter affecting the framework’s performance. To determine its optimal value, a controlled experiment was conducted to systematically test configurations with $N \in \{0, 2, 5, 10, 20\}$. Specifically, the $N = 0$ setup serves as a baseline to assess the framework’s performance when relying solely on internal knowledge generation (without external document input), thus isolating the net contribution of external knowledge to retrieval effectiveness. For

gradient configurations from $N = 2$ to $N = 20$, the experiment quantifies the bidirectional effects of retrieval scale on knowledge fusion efficiency: smaller N may lead to insufficient coverage of external evidence, causing key information omission and limiting recall improvements; larger N (e.g., $N = 20$) risks introducing noise from redundant documents, which in turn can disrupt truth discrimination during knowledge validation and dilute the ranking positions of relevant documents, thereby affecting the framework's performance. Notably, the HotpotQA dataset was chosen for this experiment due to its multi-hop reasoning nature, which imposes strict requirements on cross-document evidence integration and can sensitively reflect the framework's knowledge verification efficiency under different configurations.

When $N = 0$, the framework operates solely on validating the original triple set without external knowledge calibration. As shown in Figure 4, the lack of reference documents for factual verification of internally generated knowledge results in measurable 3.0% retrieval performance degradation compared to baseline systems. This finding reveals two fundamental insights: (1) the intrinsic hallucination risks in autonomous knowledge generation processes; (2) the essential requirement of external knowledge verification for robust system operation. Through parameter sweep analysis, peak $N = 2$ with subsequent monotonic degradation as N increases to 20. Technical evaluation indicates that at $N = 2$, pre-retrieved documents facilitate semantic space reconstruction via limited authentic data samples, effectively boosting query expansion precision. Beyond this threshold, however, cumulative noisy document influx and context window expansion cumulatively degrade the model's discriminative capacity for valuable triple updates. This non-linear relationship between reference document quantity and retrieval efficacy demonstrates that document overabundance surpasses the architecture's context processing limits, inducing critical evidence dilution through irrelevant information interference[39]. Future work should investigate preprocessing techniques including document length normalization and semantic distillation to enhance input quality during knowledge validation phases.

4.7 Large Language Model Capability

The core architecture of this framework is tightly integrated with LLM components. Given computational resource limitations, we selected Qwen-2-7B as the baseline architecture. To systematically evaluate the framework's performance boundaries, controlled experiments were conducted to measure the correlation between model capacity and retrieval efficacy. This study therefore adopts three standard evaluation metrics (Recall@k, MRR, and MAP) for rigorous performance quantification. Our experimental protocol includes comparative analysis of Qwen-2 series configurations (7B, 57B, and 72B parameters) under identical test conditions. By isolating the critical parameter of LLM scale, we examine whether the framework maintains proportional performance improvements with increasing model capacity. To assess cross-platform compatibility, the experimental design incorporates the state-of-the-art Deepseek-V3 model[23] as a cross-architecture reference. The empirical results reveal two principal findings: (1) architectural independence from specific LLM implementations, and (2) consistent

compatibility across heterogeneous model architectures. These findings provide valuable empirical guidance for model selection in production deployment scenarios.

Experimental results reveal a statistically significant positive correlation between LLM parameter scale and framework performance: As the model scale increased from Qwen-2-7B to Qwen-2-72B, the framework exhibited consistent performance gains across comprehensive evaluation metrics. When integrating higher capacity LLMs, the knowledge integration and verification mechanism effectively leverages model potential, achieving enhanced document ranking accuracy in complex retrieval scenarios. The strong empirical performance with Deepseek-V3 confirms the architecture-agnostic nature of the framework and the cross-model compatibility. Quantitative analyses show the framework's capabilities are primarily constrained by the base LLM's hallucination propensity, with scaling effects reducing hallucination rates while improving instruction-following capacity. In particular, given resource constraints that require primary experiments with Qwen-2-7B, the current results suggest substantial potential for further enhancement of capabilities through the deployment of larger-scale models.

4.8 Ablation Study

This study conducts systematic ablation experiments on the core components of the KUG framework to quantitatively evaluate the contribution of each module to model performance. The experiments focus on three key dimensions:

- (1) **w/o Info**: Investigate the role of internal knowledge in retrieving external information;
- (2) **w/o Docs**: To verify the effect of external knowledge in the update phase, set $N=0$ in the experiment;
- (3) **w/o CA(Complementary Augmentation)**: To explore the role of information augmentation strategies in the update phase, update only internal information without acquiring additional information from external knowledge during the experiment;
- (4) **w/o Triplet**: To evaluate the value of triplet-structured representation for knowledge integration, do not generate triplets but organize knowledge only in document form during the experiment.

Through systematic ablation of individual components, we establish controlled experimental groups and conduct comparative performance analysis on the HotpotQA dataset. The experimental design follows the single-variable principle to ensure rigorous comparisons with controlled single-variable conditions, offering empirical validation of each framework module's functional efficacy while maintaining methodological consistency across all test scenarios.

Internal Knowledge in Retrieving: As shown in Table 2, during external knowledge retrieval, directly utilizing raw queries without internal knowledge guidance results in measurable performance degradation. Analytical evidence indicates that retrieval processes lacking internal knowledge guidance yield external information exhibiting limited semantic overlap with the model's intrinsic knowledge base, which impedes effective hallucination correction during the knowledge integration phase and ultimately induces error propagation through successive processing stages. This cascading effect quantitatively demonstrates the critical role of our proposed

Table 2: External knowledge updating significantly mitigates hallucinations from internal parametric knowledge.

Method	Recall@40	MRR	MAP
w/o	0.95	0.6541	0.8403
Info	(-0.7%)	(-0.1%)	(-0.8%)
w/o	0.9425	0.6246	0.8046
Docs	(-1.6%)	(-4.6%)	(-5%)
w/o	0.9125	0.5972	0.7490
CA	(-4.7%)	(-8.8%)	(-11.6%)
w/o	0.9475	0.6408	0.8290
Triplet	(-1%)	(-2.2%)	(-2.1%)
KUG(ours)	0.9575	0.6549	0.8470

knowledge-guided retrieval mechanism in maintaining information consistency across system components.

External knowledge in the update phase: In the Internal-External Knowledge Confrontation phase, this work empirically demonstrates the impact of external knowledge integration on model performance. As presented in Table 2, when configured with retrieval window size $N = 0$, the model exhibits significantly degraded retrieval performance compared to the standard retrieval baseline. This observation reveals that exclusive reliance on parametric knowledge induces severe hallucination issues. With $N = 2$ retrieved documents, all key evaluation metrics show statistically significant improvements over the knowledge-free setting. These findings not only confirm the essential role of external knowledge grounding but also expose fundamental limitations of parametric knowledge-only approaches in open-domain scenarios—internal representations unanchored by external evidence demonstrate heightened hallucination tendencies, ultimately leading to suboptimal retrieval outcomes.

Complementary Augmentation: As shown in Table 2, results from the Complementary Augmentation method demonstrate that incorporating external beneficial documents during knowledge updates yields statistically significant improvements over basic update strategies, with the Recall@40 metric showing a 1.6% increase and ranking-sensitive metrics achieving approximately 5% gains. These findings suggest that optimizing language models’ knowledge systems necessitates not only routine updates but strategic integration of external knowledge sources. Further analysis reveals that LLMs’ inherent corpus interaction limitations create representational gaps in unsupervised scenarios, resulting in suboptimal context-aware retrieval performance. This empirical evidence confirms the necessity of external knowledge guidance during iterative model updates, highlighting its critical role in maintaining domain-specific reasoning capabilities.

Triplet-based Information Organization Schema: The ablation study results in Table 2 reveal that converting raw text into triplet-based representations yields a minimum 1.0% improvement on the base recall metric, with gains surpassing 2.0% on ranking-based evaluation metrics (MAP, MRR). These findings indicate that explicit structural representation of knowledge through triplets

enables more efficient processing of intricate semantic relationships and information integration, consequently improving both precision and logical consistency in generated responses.

Based on systematic evaluation of experimental outcomes, the ablation studies systematically demonstrate the critical importance of individual components in the KUG framework, with each module contributing to performance enhancement, robustness improvement, and knowledge constraint mitigation. These findings establish a solid theoretical foundation for subsequent framework optimization and scalability enhancements while maintaining coherent architectural integrity.

5 Conclusion

This study presents a novel retrieval method, KUG, which operates without the need for training. KUG is built upon a collaborative architecture that integrates LLMs with retrieval systems, effectively merging the internal knowledge parameters generated by the model with the authentic information from external knowledge bases. This approach capitalizes on the complementary advantages of both information sources while circumventing the complexities associated with traditional training processes. Experimental evaluations demonstrate that the KUG framework successfully integrates the semantic generation capabilities of LLMs with the factual verification functions of external retrieval systems, achieving a significant improvement in result accuracy and reliability while maintaining retrieval efficiency.

6 Limitations and Future

The current research is constrained by resource limitations, and the proposed method has not achieved optimal performance in several key technical aspects, specifically manifested as follows: First, no prompt engineering optimization was conducted, resulting in insufficient depth in mining the knowledge within the model’s internal parameters and underutilization of query generation capabilities. Second, to highlight the core idea of the framework and simplify the research process, large language models were employed in the construction and integration of dynamic knowledge graphs, leading to relatively high computational costs. Future research can be expanded in the following directions: first, conducting prompt engineering optimization to enhance model capabilities; second, addressing the high costs associated with graph operations by developing heuristic algorithm-based knowledge replacement and update strategies to improve operational efficiency while ensuring system interpretability. These research directions will provide more universal technical solutions for the engineering implementation of retrieval-augmented generation frameworks.

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Use of GenAI

The usage scenarios of GenAI in this paper are as follows: data generation in the FIQA-GR dataset, article polishing, and grammar checking.

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